

3. Determine the equation of a quadratic function that has an x-intercept of 4, and passes through the point (3, -5).

$$y = a(x-4)^2 + b \rightarrow a = -5$$

$$= 5 = a(3-4)^2$$

$$y = -5(x-4)^2$$

4. Determine the vertex of this quadratic function $f(x) = 5x^2 + 7x - 6$

$$f(x) = (5x-3)(x+2)$$

roots at $\frac{3}{5}$ and -2

$$AOS = \frac{\frac{3}{5} - 2}{2} = -\frac{7}{10}$$

$$f(-\frac{7}{10}) = 5(-\frac{7}{10})^2 + 7(-\frac{7}{10}) - 6$$

$$= -8.45$$

COMMUNICATION

1. A quartic function can have three zeros. Explain how to use this fact to graph the function.
Draw a sketch to support your explanation.

It can have a root of order 2 and have the graph

(2) that it goes from 2-2 because

APPLICATION

1. Determine an equation in factored form for a polynomial function with zeros 0, 2 (order 2), roots, and 5 passing through a point (6, 32).

$$y = a x (x-2)^2 (x-5) \rightarrow a = \frac{1}{3}$$

$$32 = a(6)(6-2)^2(6-5)$$

$$32 = 96a$$

$$y = \frac{1}{3} x (x-2)^2 (x-5)$$

2. The table of values represents a polynomial function. Determine the value of the leading coefficient.

x	y
-3	612.5
-2	336
-1	157.5
0	56
1	10.5
2	0
3	3.5
4	0

3 - degree 3
↳ third differences are constant

$$a_n n! = CFD$$

$$a_n (3!) = (-21)$$

$$a_n = -\frac{21}{6}$$

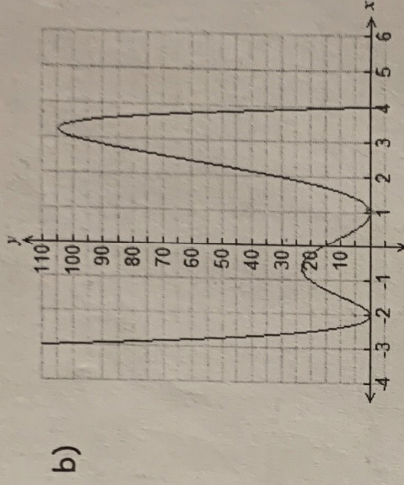
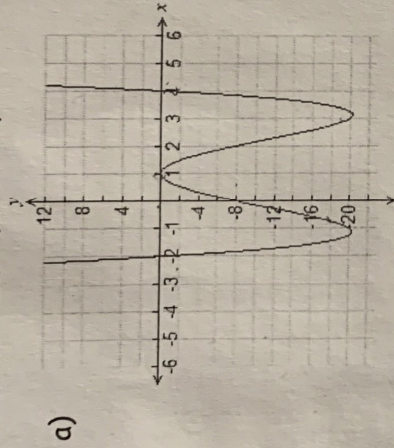
∴ the leading coefficient is $-\frac{21}{6}$

Total

16 / 16

KNOWLEDGE

1. Given the following graphs, complete the chart below:

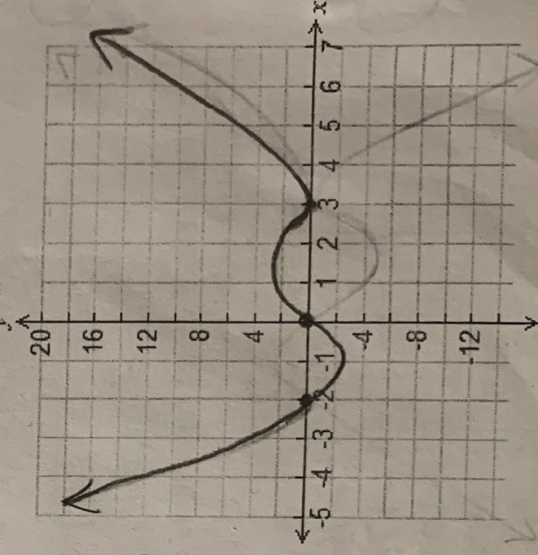


(0.25 marks each)

Graphs	Odd/Even Degree	Least Possible Degree	End Behaviour	# absolute minimums	# of local maximums
a)	even	4	$x \rightarrow -\infty, y \rightarrow \infty$ $x \rightarrow \infty, y \rightarrow \infty$	1	1
b)	odd	5	$x \rightarrow -\infty, y \rightarrow \infty$ $x \rightarrow \infty, y \rightarrow -\infty$	0	2

2. Given the function
- $f(x) = 3x(x-3)^2(x+2)$
- , graph the equation clearly showing all intercepts

and fill in the interval chart:



Interval	$(-\infty, -2)$	$(-2, 0)$	$(0, 3)$	$(3, \infty)$
Sign of $f(x)$	+	-	+	+